جامعة خليفة
Khalifa University

PERSONAL PROTECTIVE EQUIPMENT

Copies of this document available in KU EHS Intranet are only the latest, valid and controlled documents. All other copies are uncontrolled and valid only on the date printed. Last printed: 15/6/2020 3:58:00 PM

جامعة خليفة
Khalifa University

Title of Document : Personal Protective Equipment
Document No. : KU – FD – EHS – MPM – 06

Issue No. : 01
Issue Date : 19/11/2012

Revision No. : 01
Revision Date : 07/06/2018

Activity	Name / Designation	Signature	Date
Prepared by	Roberto Macahig EHS Inspector		04/03/2018
Reviewed / Updated by	Fadi Al Jallad EHSS Compliance		09/04/2018
Endorsed for Approval by	Belal Irshaid EHSS Manager		15/05/2018
Approval & Issue by	Hamza Kazim Facilities, Administration & EHSS Vice President		07/06/2018

Copies of this document available in KU EHS Intranet are only the latest, valid and controlled documents. All other copies are uncontrolled and valid only on the date printed. Last printed: 15/6/2020 3:58:00 PM

جامعة خليفة
Khalifa University

CONTENTS

1. PURPOSE: 4
2. SCOPE: 4
3. RESPONSIBILITIES: 4
4. ABBREVIATION / DEFINITIONS: 6
5. PROCEDURE: 7
5.1 PROCESS: 8
5.2 SIGNAGE: 9
5.3 PPE SELECTION: 9
5.4 CORRECT USE OF PPE: 20
6. REFERENCE STANDARDS: 21
7. ANNEXURE: 21
8. RECORDS / DOCUMENTS / FORMATS: 21

Copies of this document available in KU EHS Intranet are only the latest, valid and controlled documents. All other copies are uncontrolled and valid only on the date printed. Last printed: 15/6/2020 3:58:00 PM

1. Purpose:

The purpose of this Guideline is to provide information in the selection, use and maintenance of personal protective equipment, including protective clothing, wherever it has been identified as a risk control measure. All hazards and risks in a laboratory must be identified by a risk assessment and eliminated or controlled by the use of the Hierarchy of Risk Controls, including the use of personal protective equipment.

The purposes of this procedure are:

- To establish personal protective equipment requirements compliance and set specifications for the use and control of personal protective equipment for all KU operations and activities.
- Provide a safe and healthy environment for employees of KU;
- Provide guidance on a range of Personal Protective Equipment and on how to protect employees as a last line of defense where a risk cannot be adequately controlled by other measures.
- To define the uniform needs of the municipality employees & contractors working in KU building and premises and the area where uniform shall be worn.

2. Scope:

This procedure applies to all staff, students and contractors who undertake any activity on behalf of KU including mechanical, chemical and biological laboratories, and when on field trips. This procedure does not apply to designated computer laboratories.

3. Responsibilities:

3.1 EHS Steering Committee Representative: Responsible for

- Implement, maintain and update this procedure as required.
- Ensure necessary recourses are available for an effective implementation and maintenance of this procedure.
- Ensure compliance with the relevant local and national legislative requirements.

3.2 EHS Manager: Responsible for

- Ensuring the awareness of this procedure throughout KU.
- Issue and maintain this procedure.
- Approval of this procedure.

Copies of this document available in KU EHS Intranet are only the latest, valid and controlled documents. All other copies are uncontrolled and valid only on the date printed. Last printed: 15/6/2020 3:58:00 PM

3.3 EHS Coordinators: Responsible for

- Shall ensure that internal EHSMS objectives and targets are set and are promoted across the Municipality.

3.4 Facility Management Department – EHS Section:

To assist areas in the implementation of these guidelines the EHS Section will advise on the type of PPE available and in conjunction with the area, provide assistance with the selection, inspection, maintenance or training required.

- Communicating this procedure to all levels of KU.
- Responsible for the overall performance of this procedure.
- Assist section managers and contractors to assess the personal protective equipment needs and requirements for all job activities, and ensure that the proper protective equipment is selected.
- Periodically monitor job activities to ensure that supervisors and employees are complying with the requirements of this procedure.

3.5 Supervisors and Managers:

All staff that is designated as supervisors, area managers or laboratory managers is responsible to ensure staff, students and visitors:

- Are supplied with the appropriate PPE for their work activities.
- Adhere to the requirements of this guideline.
- Ensure that risk assessments or safe work procedures are completed that outline the need for PPE and that other risk control measures have been considered.
- Provide and make financial allocations for PPE.
- Ensure adequate training and continuing supervision are available at all times for workers, students and visitors using protective equipment.
- Ensure procedures are adopted for the selection, inspection, maintenance and cleaning of all PPE and clothing purchased by the area
- Ensure that clear and appropriate signs are positioned in locations where PPE and clothing must be worn.
- Ensure teaching employees inform and monitor the use of appropriate clothing and protective equipment by students under their supervision.

3.6 Purchasing and Procurement:

- Approve the use of any personal protective equipment before it is purchased as per requirement.
- Use protective equipment as required.

3.7 Inspectors:

- Monitoring of the effectiveness of this procedure.
- Reporting non-conformities to EHS Section.

3.8 All KU Employees or and Users of PPE:

- All staff, students and visitors must use and look after the PPE that has been identified as appropriate PPE for their activities.
- Ensure compliance with this procedure requirement in accordance with their responsibilities.
- Report all incidents, whether the injury is sustained or not.
- Report all unsafe practices or potential hazards. And anything that is hazardous to the environment.

4. Abbreviation / Definitions:

EHSSC	EHS Management Steering Committee Representatives
EHSMS	Environment, Health and Safety Management System
MSM	Management System Manual
MPM	Management Procedure Manual
ESMA	Emirates Authority For Standardization & Metrology
ASTM	American Standard Testing Methods
NFPA	National Fire Protection Association
ANSI	American National Standard Institute

Hazard: A source or a situation with a potential for harm in terms of human injury or ill-health, damage to property, damage to the environment, or a combination of these.

Approved Personal Protective Equipment:: Equipment which has been approved as meeting minimum acceptable safety standards, as given in this procedure.

Hierarchy of Risk Controls: The method of addressing and implementing risk control measures in order of importance, which is:

1. Elimination: Remove the hazard.
2. Substitution: Replace with a less hazardous substance or activity.
3. Engineering: Example installs guards on machinery, provides fume extraction.
4. Administration: Signs, Procedures, Following safe work practices, Training.
5. Personal protective equipment: Respirators, ear plugs, etc.

Operating Area: It includes all areas of Operating Facilities where a permit is required for carrying out any Construction & maintenance works. It includes but not limited to all KU premises, Site locations, and other operational units.

EHS Coordinators: Are nominated employees responsible for representing their Departments / Sections within EHSMS on matters relating to EHS.

Shall: Signifies mandatory requirements

Should: Signifies recommended / optional requirements

Risk: (In relation to any potential injury or harm) the likelihood and consequence of that injury or harm occurring.

Personal Protective Equipment (PPE): All equipment (including clothing affording protection against the weather) which is intended to be worn or held by a person at work and which protects him against one or more risks to his health or safety', e.g. safety helmets, gloves, eye protection, high visibility clothing, safety footwear and safety harnesses.

5. Procedure:

The use of PPE is supported by instruction, training, and supervision of use, regular maintenance and replacement procedures as outlined through this document.

- Special protective clothing or equipment shall be free of charge to the employee and available also to the students.
- Employees and students shall receive appropriate instruction in the use of PPE and clothing.
- Faculty and Divisional Heads shall provide supervision to ensure, as far as practicable, the appropriate equipment and clothing is being used in the manner required.
- Employees shall use protective clothing and equipment for the purpose provided and in the manner required by the University, and shall immediately inform management of any defects or deficiencies of which they become aware, in such equipment.

The Khalifa University requires all staff, students, contractors and visitors working in laboratories and preparing for field trips to risk assess their activities. Where that risk assessment indicates that hazards and risks occur which cannot be completely eliminated or controlled by application of the Hierarchy of Risk Controls, then PPE must be used.

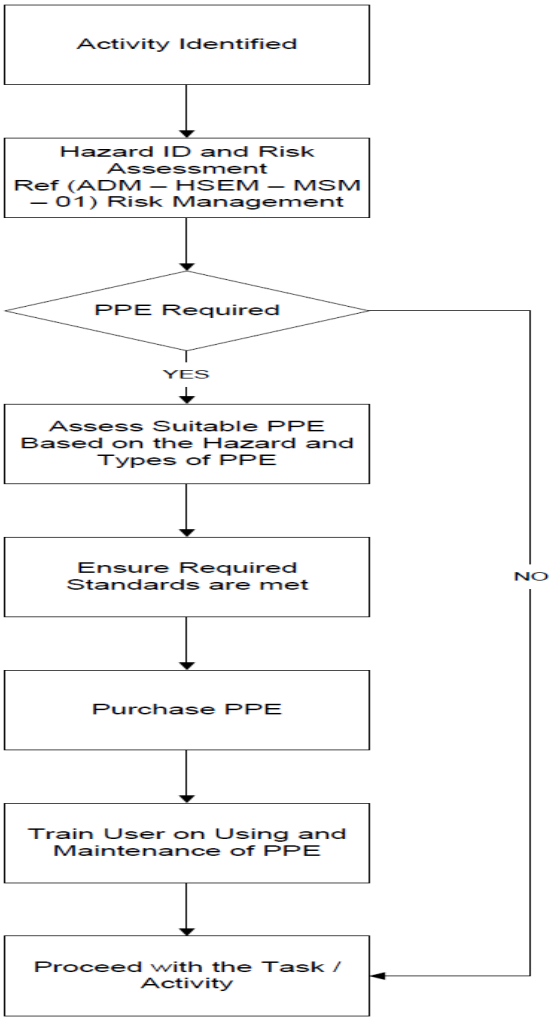
Copies of this document available in KU EHS Intranet are only the latest, valid and controlled documents. All other copies are uncontrolled and valid only on the date printed. Last printed: 15/6/2020 3:58:00 PM

All risk assessments shall take into account the minimum PPE requirements of the OSHAD-SF. These are:

- Laboratory clothing
- Protective Eyewear
- Gloves
- Closed shoes
- Hearing protection
- Additional or specialized PPE, e.g. respirators

All possible routes of entry of the hazard into the body must be taken into account when considering PPE intended to control the exposure.

5.1 Process:



5.2 Signage:

Where PPE is expected to be worn inside a facility, there needs to be appropriate PPE Signage on the outside of the doors leading into the facility. PPE Signs are a tool that Laboratory Managers can use to help achieve this.

5.3 PPE Selection:

PPE shall be selected, used and maintained in accordance with the relevant AD EHSMS Code of Practices and any relevant international standards. E.g. some personal protective equipment Australian Standards include:

- AS/NZS 1336 Recommended practices for occupation eye protection
- AS/NZS 2161 Occupation protective gloves
- AS/NZS 2210 Occupation protective footwear
- AS/NZS 1270 Acoustics – hearing protectors
- AS/NZS 1715 Selection, use and maintenance of respiratory protective devices

Proof of compliance with the relevant Australian Standard is a prerequisite for purchase of any personal protective equipment.

Personal protective equipment must be:

- Properly selected for the individual and the task ;
- Readily available if supplied by KU;
- Clean and functional;
- Correctly used;
- Appropriately stored and maintained

When selecting the appropriate PPE consideration should be given to elements such as:

- Durability.
- Permeability.
- Abrasion resistance.
- Burn resistance.
- Insulation.
- Cut resistance.
- Ability to be cleaned.

- Size.
- Comfort.

5.3.1 Laboratory clothing:

All laboratory users shall use protective clothing appropriate to the tasks being undertaken. These may include long-sleeved laboratory coats, wraparound gowns, disposable gowns and boiler suits. Cotton or cotton/polyester is the preferred material.

Specialized clothing may be required for certain hazardous processes. Such additional clothing will be identified on the Material Safety Data Sheet (MSDS) for that workplace substance. For instance, when handling hydrofluoric acid, always wear a laboratory coat with a chemical splash apron made out of natural rubber, neoprene, or Viton.

The most appropriate type of protective clothing required for a laboratory will depend on the activities being undertaken and the associated risks. Work that predominantly involves biological risks, such as working with microorganisms, bio-fluids (blood, urine feces, sputum), viruses etc, require the use of a rear-fastening, wrap around laboratory gown. A gown has longer sleeves enabling the cuff of the gloves to be pulled over the cuff of the gown, thereby protecting the wrist from exposure to biological material. The gown also provides better protection for the upper body if a lot of work is carried out in a seated position. Where work predominantly involves the use of chemicals, particularly corrosives (acids and bases) and flammable materials, then the laboratory coat, which is generally made of a stronger cotton material, offers better protection and complies with the requirement of being able to be removed easily in the event of a chemical splash/spill.

In addition, factors such as the ability of the garment to get caught up in equipment should be considered. For example, the open end of the sleeves on a laboratory coat can get caught or knock things over whilst the baggy sleeves of a gown can get caught when reaching across equipment or other items.

It is important to take into account whether the work will be conducted in a seated or a standing position. When working with chemicals in a seated position as an additional layer of protection may be required such as that offered by a chemical resistant apron worn over the laboratory coat.

The need to protect the gap between where a glove ends and a laboratory sleeve begins should also be considered. This may not automatically result in a gown being a better choice than a laboratory coat. It may also be acceptable to use protective sleeves over the arms of the lab coat.

In summary, the choice of protective clothing to be worn in a particular laboratory is based on risk and thus should be determined in consultation the literature and with other laboratory users through a documented risk assessment. Whichever option is taken it is important to communicate the need to wear the clothing as intended i.e. gowns properly fastened at both ties and laboratory coats fully buttoned up.

Circumstances in which PPE may be required to be worn include:
Head protection (hard hats, helmets, sun hats) shall be provided where there is a risk of injury to the head, such as a person struck on the head by a falling object, a person may strike his/her head against a fixed object, there may be inadvertent head contact with electrical hazards.

Eye protection (goggles, safety glasses) shall be provided where a risk of eye injury exists. Typical hazards might include flying particles, dust, splashing substances, harmful gases, vapors, aerosols, and high intensity radiation from welding operations.

Hearing protection (ear plugs, ear muffs) shall be provided where a risk of noise induced hearing loss exists. The need for hearing protection may be assessed through noise monitoring or surveys.

Respiratory protection (respirators, face masks, cartridge filters) shall be provided where there is a risk of airborne contaminates. This will minimize the risk to of exposure to an atmosphere that is or may be injurious to health.

Hand protection (gloves, gauntlets) shall be provided where there is an identified hazard associated with a potential for a hand injury. A list of hazards shall be compiled for each workplace and suitable hand protection obtained to minimize risk.

Protective footwear (safety boots, gumboots, enclosed shoes) shall be provided where the nature of the work exposes the employee to a medium to high risk of injury to feet, e.g. occupations such as workshop/maintenance and gardening staff.

Body protection: (high-visibility garments, thermal wear, overalls, aprons, safety harnesses) shall be provided to minimize risk of injury occurring in the body. Examples may include those who are required to work outdoors and are exposed to the sun's rays for continuous periods in a day. Direct exposure of the skin to UV radiation from outdoor work shall be minimized by providing hats, long sleeves/trousers and an adequate supply of sunscreen.

Outerwear: (high visibility safety vests, reflective vests, Fluoro jackets) shall be provided to highlight the worker in the area often used where there is a risk of injury associated with working on or near roadways or near moving traffic or moving plant.

The following table provides a summary of protective clothing types

Protective Clothing Type	Advantages	Disadvantages
Laboratory Coat:	• Offers more protection in the event of a fire • Offers better protection against corrosives • Comes in a range of sizes • Is able to be removed quickly in the event of an emergency	• Gaping sleeves may expose the wrist • The open sleeve may get caught up in the equipment or knock things over • Exposes the top of the upper legs and lap when working in a seated position • Exposes the neck • Designed to be used whilst standing • Offers spatter protection only and a further layer of protection such as chemical apron is required for direct spills
Laboratory Gown	• Offers better protection against biological risks (covers full front of body) • Tight cuffs fully protect the wrist • Offers better protection for the lap and upper legs • The snug fitting sleeve ends prevents the sleeve getting caught up in equipment/knocking things over.	• Offers poor protection against corrosives • Offers poor fire resistance • Has limited size choices
Chemical Splash Apron	• Can be worn easily over a lab coat to provide an extra layer of protection against strong corrosives and other dangerous chemicals.	• They cannot be used on their own as they do not provide protection to the arms and upper torso.
Dust coats	• Keeps street clothing clean in dusty areas	

Laboratory clothing must be removed before leaving the laboratory to reduce the risk of contamination to non-laboratory users. Where dust coats are used to simply keep street clothing clean and no risk of contamination exists (i.e. No

chemical, biological or radiation hazards exist), then the dust coat does not need to be removed. However to avoid confusion the dust coat should be distinguishable from lab coats or gowns (e.g. Use a colored dust coat).

The University provides a laundry service for laboratory clothing for staff. Where students are laundering their own laboratory clothing, they should use a suitable detergent (e.g. for biological contamination Napisan may be appropriate) and launder separately to other clothes.

5.3.2 Protective Eyewear:

All laboratory users shall use protective Eyewear where there is a risk of damage to the eyes. Sources of damage may include splashing of liquids hazardous to the eyes, impact, and foreign particle entry and laser radiation.

Safety Glasses:

Safety glasses have lenses that are impact resistant and frames that are much stronger than standard prescription glasses. Safety glasses must have side shields and must be worn whenever there is a possibility of objects striking the eye, such as particles, glass or metal shards.

Safety glasses may not always provide adequate protection from chemical splashes as they do not seal to the face. Safety glasses may be adequate where the potential splash is minimal e.g. opening Eppendorf tubes, or where the chemicals in use are of low toxicity.

Ordinary prescription glasses do not provide adequate protection from injury to the eyes and could even be hazardous to the wearer. For further information on prescription glasses, see the section below on prescription spectacles.

Splash Goggles:

The goggles come in a variety of styles for maximum comfort and splash protection. Chemical splash goggles should be worn when there is a high potential for splash from a hazardous material. For example, goggles should be worn when working with glassware under reduced or elevated pressure and when the glass apparatus is used in combustion or other high temperature operations. Like safety glasses, goggles are impact resistant. Chemical splash goggles shall have indirect ventilation so hazardous substances cannot drain into the eye area. Some can be worn over prescription glasses.

Face shields:

Face shields are required when working with large volumes of hazardous materials, either for protection from splash to the face or flying particles. Face shields may need to be used in conjunction with safety glasses or goggles.

AS2243.1 Safety in laboratories (Planning and Operational aspects) provides the following examples where a face shield should be used:

a) Where glass apparatus is evacuated, recharged with gas or pressurized;

b) When pouring corrosive liquids;

c) When using cryogenic fluids;

d) When combustion processes are being carried out;

e) Where there is a risk of explosion or implosion;

f) When using chemicals that can cause direct damage to the skin; and

g) When using chemicals and biological agents that can be rapidly absorbed into the body via any path e.g. through the skin, eyes or nose.

The level of protection chosen shall take into account any eye and face hazards from other work being carried out in the vicinity. For some tasks, a face shield with a brow guard, chin guard or both should be used. AS2243.3 suggests face shields should be worn when opening an autoclave.

Prescription spectacles:

Prescription spectacles (as distinct from prescription eye protectors) are generally inadequate against flying objects or particles and could even be hazardous. For persons requiring eye protection in addition to sight correction, the use of prescription spectacles worn with additional protection, e.g. over glasses, wide vision goggles or clip-ons will be necessary. It is important to note the following disadvantages of wearing these with prescription glasses:

- The majority of prescription eye protectors can provide no more than low impact protection because of their lightweight design. Where medium impact resistance is required, medium impact resistant eye protectors complying with AS/NZS 1337 suitable for use over prescription lenses shall be used over the prescription lenses.
- The use of safety goggles worn over prescription lenses will not necessarily provide protection against impact from flying objects. Fracture of the prescription lenses can occur when the safety goggles deflect under impact, even if the goggles are not penetrated.

Contact lenses:

Contact lenses are not eye protective devices and wearing them does not reduce the requirement for eye and face protection. When the work environment entails exposure to intense heat, molten metals, a highly particulate atmosphere, corrosive substances or any of the following substances (acrylonitrile, methylene

chloride, 1,2 dibromo-3-chloropropane, ethylene oxide and methylene dianiline), contact lens use should be avoided.

The following safety measures must be implemented should contact lenses be worn by individuals working with chemicals:

- Conduct a risk assessment prior to working with any chemicals or biological material to determine what type of eye protection is required, and whether the wearing of contact lenses should be avoided.
- Notify workers and visitors about any defined areas where contact lenses are restricted.
- Identify to supervisors all contact lens wearers working in chemical environments to ensure that the proper risk assessment is completed and the appropriate eye protection and first aid equipment are available.
- In the event of a chemical exposure, begin eye irrigation immediately and remove contact lenses as soon as practical. Do not delay irrigation while waiting for contact lens removal.
- Instruct workers who wear contact lenses to remove the lenses at the first sign of eye redness or irritation.

Laser eye protection:
To ensure the correct laser eye protection is selected users must understand:

- What type of laser is to be used;
- What power and wavelength is the laser;
- Whether impact resistance is required.

The local Radiation Safety Officer should be consulted when selecting a laser eye protection.

5.3.3 Gloves:

Many laboratory activities could lead to contamination of the hands and therefore potentially serious injury. A risk assessment should identify the type of glove required for the chemical and activity contemplated and the thickness of the material. Disposable gloves will only protect against splashes. If you know that you will be getting your hands wet, you may need thicker gloves.

It is important to choose gloves carefully as many hazardous substances pass readily through some glove types. The MSDS for the chemical will indicate the appropriate glove material. There are also glove selection services and permeation guides on the internet.

You must use the MSDS from the /manufacturer supplier of the chemical and you must use the website for the manufacturer of the gloves that you are purchasing.

Gloves must be removed before leaving the laboratory to reduce the risk of contamination to non-laboratory users.

5.3.4 Protective footwear:

All laboratory users must wear footwear appropriate to the hazards of the laboratory. As a minimum, enclosed footwear must be worn. Shoes made of absorbent material or woven fabric e.g. most types of joggers, runners and tennis shoes), should not be worn in laboratories where exposure to corrosive or chemicals that are easily absorbable through the skin could occur.

5.3.5 Hearing protection:

All laboratory users must wear appropriate hearing protection when noise may damage or impair hearing, or as directed by legislation.

KU is required to eliminate or minimize any noise which may damage or impair hearing. New South Wales legislation defines a workplace as unsafe if

Any person is exposed to noise levels that exceed an 8 hour noise equivalent of 85dB (A) or that peak at more than 140dB (C).

In determining the choice of hearing protection reference should be made to Table below

CLASS OF HEARING PROTECTOR REQUIRED	
$L_{Aeq, 8h}$, dB (A)	**Class**
Less than 90	1
90 to less than 95	2
95 to less than 100	3
100 to less than 105	4
105 to less than 110	5
Greater than 110 or equal to 110	Seek specialist advice

The EHS Section provides a noise survey service upon request.

5.3.6 Respirators and dust masks:

Respirators should only be used to minimize the risk from inhalation risks (such as dusts, mists, fumes and vapors) if other hazard control methods are not practicable.

Situations at KU when they can be considered include:

- When engineering solutions such as mechanical ventilation, fume cupboards, local exhaust ventilation etc. Are not technically feasible;

- While engineering controls are being installed or repaired;
- If emergencies or other temporary situations arise (e.g. Cleaning up spills etc.).

What are the different classes of respirators?

The two main types are:

- Air-purifying respirators and;
- Supplied-air respirators.

Air Purifying Respirators:
Some Air-Purifying Respirators work by filtering out particulates (e.g., dusts, metal fumes, mists, etc.) from contaminated air. Others can purify air by adsorbing gases or vapors onto an adsorbing material (like charcoal) in a cartridge or canister. They are either negative pressure units (most common types used at KU) or positive-pressure units such as powered air-purifying respirators (PAPRs). A powered air purifying respirator requires a battery to operate. The battery functions to supply power to the impeller that draws ambient air through the filter. A powered air purifying respirator has a headpiece which can be in the form of a respirator hood, mask, a loose fitting face-piece or a full face-piece.

Negative pressure Air Purifying Respirators are tight-fitting and are available as either:

- Half-face mask (covering the face from the nose to below the chin), or
- Full face-piece (covering the face from above the eyes to below the chin).

Examples include:

- Disposable dust masks (half face only);
- Particulate respirators which have a filter for trapping particulate matter;
- Chemical cartridge respirators that can have a combination of chemical cartridges and a dust pre-filter. Different options are available such as acid gas cartridges (for example if exposure to hydrochloric acid was an issue) or organic vapor cartridges (e.g. If the issue was related to Ethanol or Xylene).

There are three classes of particulate filters suitable for filtering finely divided solid or liquid particles, or both, from inhaled air. These are classified in accordance with the tests in AZ/NZS 1716 and are classified as P1, P2 and P3.

Copies of this document available in KU EHS Intranet are only the latest, valid and controlled documents. All other copies are uncontrolled and valid only on the date printed. Last printed: 15/6/2020 3:58:00 PM

- **P1 classification** is for protection against mechanically-generated particulates. Suitable applications include woodworking, workshop and construction type activities.
- **P2 classification** is for protection against mechanical and thermal-generated particulates, mists and fumes.

Cartridges are color coded in accordance with the type of contaminant they offer protection against. E.g.:

- Dust – purple cartridge
- Solvents – black cartridge
- Formaldehyde – black cartridge
- Ammonia – green cartridge
- Acid Gas – yellow cartridge

- **Class P3 respirators** are intended for use against all particulates including highly toxic materials. However for non-powered respirators, a class P3 respirator classification can only be assigned where a class P3 filter is combined with a full face piece

Air purifying respirators must not be used if the hazard is an Oxygen deficient environment.

Supplied Air Respirators:

Supplied-air respirators supply clean air from either an air filled gas cylinder i.e. Self-Contained Breathing Apparatus (SCBA) or from an external airline which supplies cleans compressed air from outside the work area.

Supplied-air respirators may have either tight-fitting or loose-fitting respiratory inlets. Respirators with tight-fitting respiratory inlets have half or full face-pieces. Types with loose-fitting respiratory inlets can be hooded or helmets that cover the head and neck. These respirators are used where there is a potential for an Oxygen deficient environment or if there is a concentration of toxic gas which is immediately dangerous to health.

Points to consider before choosing a respirator:

- A risk assessment should be conducted to identify the hazardous substances that present the inhalation risk. The hierarchy of risk controls should have been implemented to reduce the risk as much as possible before applying the control of PPE.

Copies of this document available in KU EHS Intranet are only the latest, valid and controlled documents. All other copies are uncontrolled and valid only on the date printed. Last printed: 15/6/2020 3:58:00 PM

- An attempt should be made to quantify the potential exposure from the hazardous substances - this can be done provisionally using simple and inexpensive gas detection systems such as a pump and calorimetric tubes (contact your EHS coordinator).
- Determine whether a half face or full face respirator is required. Once the appropriate respirator and cartridges are selected a fit test must be conducted. The wearer may be required to obtain medical clearance if the duration of use is frequent or of significant duration (e.g. May not be required if the use is for a one off cleanup of a chemical spill which may only require wearing it for minutes duration).
- Factors which influence the effectiveness of the seal of the respirator onto the face must be eliminated e.g. no beards, long sideburns or stubble.
- Potential wearers must be trained in the respirator selection, use, cleaning, storage and maintenance.

Maintaining the Respirator:

Reusable respirators must be cleaned after use. The cartridges should be removed and the respirator washed in warm soapy water, rinsed and left to dry. Solvents must never be used to clean respirators. Once dry the respirator and cartridges should be returned to a seal proof bag or container and stored in a personal locker or cupboard dedicated to the storage of personal protective equipment. The manufacturer's instructions will provide further details on maintenance which can be incorporated into the Safe work procedure for the use of the respirators.

Any defects such as scratches, loose parts, tears, holes etc. affect the proper functioning of the respirators and such respirators should not be used.

Use of the Cartridges should be tracked/ logged to ensure they remain effective. There is no absolute way of determining whether or not they are still effective; however as an example if they are being used to protect against a material that presents an odor, as soon as the odor starts to become noticeable whilst wearing the respirators is confirmation that the cartridges are no longer effective. Often it is easier to use a blanket rule such as: replace 6 months after first use. In any case they must never be used past their expiry date.

5.4 Correct Use of PPE:

5.4.1 Fitting:

The correct fit is a prerequisite for the correct operation of PPE and must be checked before the PPE is used. This is especially important for respiratory devices where a good facial seal is required.

Instruction and training:

Staff, students and visitors must be taught the correct way to use the PPE. Instruction should include the need for the equipment, its basic design principles, its application and limitations.

Annexure 2 outlines the minimum training requirement required for the relevant PPE. In some cases the use of PPE is self-explanatory e.g. wearing glasses, dust masks or hearing protection. In other situations where the training equipment is more complicated it is required for the correct fit, use, selection, storage and maintenance of PPE e.g. breathing apparatus, fall arrest equipment.

When training is to be provided it shall cover:

- The method for fitting and/or wearing the PPE type.
- The intended application of the PPE.
- Information on the locations or activities where the PPE is to be worn.

Further information is provided in Annexure 2.

Maintenance:

All PPE must be maintained, tested and stored according to the manufacturer's requirements. The PPE must be kept in a clean, hygienic and effective condition. When not in use, PPE should be stored sealed, in a convenient, accessible and appropriate manner.

Laboratory coats and gowns must be hung on coat hooks provided and not draped over the back of laboratory chairs.

Issuing PPE:

All staff, research students and visitors are to be issued with PPE when required. The only exception applies to research students where the provision of steel-capped safety boots is their responsibility.

Undergraduate students are required to provide their own PPE, such as laboratory coats and protective Eyewear.

Copies of this document available in KU EHS Intranet are only the latest, valid and controlled documents. All other copies are uncontrolled and valid only on the date printed. Last printed: 15/6/2020 3:58:00 PM

5.4.2 Review and evaluation:

Risk assessments:

Risk assessments must be periodically reviewed to ensure that all hazards and risks in the laboratory have been identified and eliminated, or controlled by the use of the Hierarchy of Risk Controls.

PPE:

The adequacy of personal protective equipment must be regularly assessed to ensure that personal injuries are not occurring.

6. Reference Standards:

6.1 ISO 14001:2015 Environmental Management System – Requirements.
6.2 ISO 45001:2018 – Occupational Health and Safety Management Systems – Requirements.
6.3 Abu Dhabi Occupational Safety and Health System Framework (OSHAD-SF) – CoP 2.0 – Personal Protective Equipment.
6.4 Abu Dhabi Occupational Safety and Health System Framework (OSHAD-SF) – CoP 3.0 – Occupational Noise.
6.5 Abu Dhabi Occupational Safety and Health System Framework (OSHAD-SF) – CoP 2.0 – Safety Signage and signals.
6.6 GSO 518/1994.
6.7 Ansell glove guide: http://www.ansell.com.au/specware/search.asp
6.8 AS/NZS 1336 Recommended practices for occupation eye protection
6.9 AS/NZS 2161 Occupation protective gloves
6.10 AS/NZS 2210 Occupation protective footwear
6.11 AS/NZS 1270 Acoustics – hearing protectors
6.12 AS/NZS 1715 Selection, use and maintenance of respiratory protective devices
6.13 AS/NZS 2243 Safety in laboratories (series)
6.14 AS/NZS 1269.3 Occupational noise management - Part 3: Hearing protector program

7. Annexure:

7.1 **Annexure 1** – Personal Protective Equipment Signs.
7.2 **Annexure 2** – Selection, Inspection, Maintenance and Training Requirements.

8. Records / Documents / Formats:

8.1 Personal Protective Equipment Checklist and Periodic Preventive Mantainance Record.
8.2 Hazard Identification Sign Order Form No.: KU – FD – EHS – FOR – 09.

Copies of this document available in KU EHS Intranet are only the latest, valid and controlled documents. All other copies are uncontrolled and valid only on the date printed. Last printed: 15/6/2020 3:58:00 PM

Annexure: 1

Personal Protective Equipment Signs

	Protective Eyewear - Goggles - must be worn. This is the most effective protective Eyewear for all work chemicals. Check the supplier for approved chemical-resistance, splash-proof goggles. **See** AS/NZS 1336 Recommended practices for occupational eye protection.
	Face Shield – must be worn. This would normally be used for decanting tasks including decanting of liquid Nitrogen. To protect eyes, approved safety glasses or goggles must also be worn with this device. **See** AS/NZS 1336 Recommended practices for occupational eye protection.
	Safety Glasses must be worn. This is a minimal standard of eye protection and does not protect against all eye hazards. There may be insufficient top and side protection to fully protect the eyes in which case goggles must be worn. **See** AS/NZS 1336 Recommended practices for occupational eye protection.
	Foot Protection – fully enclosed footwear must be worn. No open toed or heeled footwear. Absolutely no thongs or sandals. **See** AS/NZS 2210 Occupation protective footwear
	Hand Protection – Gloves offering appropriate protection to the hazard must be worn. Gloves may be disposable (e.g. Nitrile) or re-useable e.g. fully PVA, PVC or Nitrile. Consult the MSDS for further information. **See AS/NZS 2161** Occupation protection gloves

Apron	**An Apron** made of material resistant to the hazardous material must be worn. Consult the MSDS for information on the type of protective clothing required.
Lab. Coat	**Lab Coats must be worn in this area.** This would be a typical PPE to be worn in any laboratory where chemicals are being used.
Overalls	**Lab Gown must be worn in this area.** This would be typical PPE to be worn in any PC2 laboratory where microorganism and/or cells are being used and grown Gowns with cuffs must be worn when working in a biological safety cabinet.
Mask	**A Dust mask is required** because there is a risk of inhaling vapors, dust or mists. These masks are disposable. **See** AS/NZS 1715 Selection, use and maintenance of respiratory protective devices.
Respirator	**An Air Purifying Respirator is required.** These respirators require chemical cartridges to be attached. Cartridges need to be replaced regularly depending on usage. Check the MSDS for the chemicals to select the most appropriate cartridge. **See** AS/NZS 1715 Selection, use and maintenance of respiratory protective devices.

Appendix 1:

Selection, Inspection, Maintenance and Training Requirements

PPE Type	Selection	Inspection Frequency	Maintenance and Storage Requirements	Applicable legislation, Codes of Practice or Standards	Minimum Training or Testing Requirements
Clothing					
For general advice regarding occupational protective clothing, refer to AS/NZS 4501.1 for guidance.					
High visibility safety garments	Selection recommendations outlined in *Appendix C* of AS/NZS 4602.1	Visual between 3 – 6 months	Replace when necessary if the garment is badly damaged, soiled, faded or the reflective material has ceased to function. No specific storage requirements other than those listed in 5.4.	AS/NZS 4602.1	N/A
Protective clothing for users of handheld chain saws	N/A	Visual - Regular intervals when in storage and after each time of use	The manufacturer's or supplier's recommendations regarding repair, maintenance, and cleaning of the protective clothing should be observed. All broken or defective fasteners or straps should be	AS/NZS 4453.3	N/A

PPE Type	Selection	Inspection Frequency	Maintenance and Storage Requirements	Applicable legislation, Codes of Practice or Standards	Minimum Training or Testing Requirements
			repaired before use. Where the protective materials have been damaged or cut by a saw chain, the garment shall be discarded. No specific storage requirements other than those listed in 5.4.		
Clothing for protection against heat and flame	Selection recommendations outlined in *Annex C* of AS/NZS ISO 2801	Visual - Routine examinations before and after each use	Routine examinations should include checks for: a) Soiling b) Contamination from hazardous materials, including biological agents c) Physical damage (rips, tears, cuts, missing hardware and closing systems) d) Thermal damage (charring, burn holes, melting, change in color) e) Damaged or missing accessories (reflective trimming, labelling)	AS/NZS ISO 2801	N/A

PPE Type	Selection	Inspection Frequency	Maintenance and Storage Requirements	Applicable legislation, Codes of Practice or Standards	Minimum Training or Testing Requirements
			f) On-going evaluation of system fit and interfaces/ overlaps No specific storage requirements other than those listed in 5.4.		
Eye Protection					
Mesh eye and face protectors for occupational applications	N/A	N/A	N/A No specific storage requirements other than those listed in 5.4.	AS/NZS 1337.2	Fittest as outlined in *Appendix A* of AS/NZS 1337.2
Filters and eye protectors against laser radiation (laser eye protectors)	Selection recommendations outlined in *Annex B* AS/NZS of 1337.4	N/A	N/A No specific storage requirements other than those listed in 5.4.	AS/NZS 1337.4	N/A
Eye protectors for adjustment work on lasers and laser systems (laser adjustment eye-protectors)	Selection recommendations outlined in *Annex B* AS/NZS of 1337.5	N/A	N/A No specific storage requirements other than those listed in 5.4.	AS/NZS 1337.5	N/A
Filters for protection against radiation generated during welding and allied	Selection recommendations outlined in *Appendix A* AS/NZS of 1338.1	N/A	On visual inspection of the viewing area, filters shall be reasonably free of pits, scratches,	AS/NZS 1338.1	N/A

PPE Type	Selection	Inspection Frequency	Maintenance and Storage Requirements	Applicable legislation, Codes of Practice or Standards	Minimum Training or Testing Requirements
operations			grayness, watermarks, bubbles, striae, local aberrations and inclusions which could impair or prevent the use of the filters for their intended purpose. Visual inspection shall be carried out under good illumination against a suitable background with the unaided eye although spectacles may be worn if required. nspection of automatic welding filters shall be carried with the filter in the light state. specific storage requirements other than those listed in 5.4.		
Filters for protection against ultraviolet radiation	Selection recommendations outlined in *Appendix A* AS/NZS of 1338.2	N/A	N/A No specific storage requirements other than those listed in 5.4.	AS/NZS 1338.2	N/A
Filters for	Selection recommendations	N/A	N/A	AS/NZS 1338.3	N/A

PPE Type	Selection	Inspection Frequency	Maintenance and Storage Requirements	Applicable legislation, Codes of Practice or Standards	Minimum Training or Testing Requirements
protection against infra-red radiation	outlined in *Appendix A* AS/NZS of 1338.3		No specific storage requirements other than those listed in 5.4.		
Footwear					
Safety, protective and occupational footwear	Selection recommendations outlined in *Section 2* of AS/NZS 2210.1	Visual - Before and after each use	If there are any cracks, breaks in the leather, exposed toecaps or similar damage that reduces the protective qualities of the footwear, the footwear should be replaced. Guidance on assessing wear outlined in *Appendix B* of AS/NZS 2210.1 specific storage requirements other than those listed in 5.4.	AS/NZS 2210	N/A
Gloves					

PPE Type	Selection	Inspection Frequency	Maintenance and Storage Requirements	Applicable legislation, Codes of Practice or Standards	Minimum Training or Testing Requirements
Occupational protective gloves	Selection recommendations outlined in *Table 1* of AS/NZS 2161.1 Refer to the Selection and Use of Gloves Guidelines for further information	Visual - Before and after each use	N/A No specific storage requirements other than those listed in 5.4.	AS/NZS 2161	Training should be provided on the appropriate use of the gloves. This will include: a) Preferred methods of glove removal b) Assessment of fit c) Cleaning d) Maintenance, and e) Inspection of gloves for signs of wear and deterioration or degradation.
Insulating gloves for electrical purposes	Care should be taken in the selection of other protective gloves, when used in conjunction with insulating gloves for electrical purposes, to ensure that they do not negate the insulation properties of the electrical insulating gloves	Visual - Before use	Recommendations for storage, issue and examination of gloves are outlined in *Appendix C* of AS 2225 No specific storage requirements other than those listed in 5.4.	AS 2225	Gloves should be pressure tested for pinholes by sealing the entry to the glove and compressing the air trapped within by rolling the glove on it. Loss of air indicates that the glove is defective. Gloves found to be defective should not be used.
Headwear					
Occupational protective helmets	User	Visual inspection at least weekly	Care and maintenance requirements outlined in *Section 3* of	AS/NZS 1800	N/A

PPE Type	Selection	Inspection Frequency	Maintenance and Storage Requirements	Applicable legislation, Codes of Practice or Standards	Minimum Training or Testing Requirements
			AS/NZS 1800. No specific storage requirements other than those listed in 5.4.		
Hearing Protection					
Acoustics – Hearing protectors	Selection recommendations outlined in *Table 2* of AS/NZS 1270	N/A	As outlined by the manufacturer No specific storage requirements other than those listed in 5.4.	AS/NZS 1270	N/A
Respiratory Protection					
Respiratory protection	Selection recommendations outlined in *Section 4* of AS/NZS 1715	Visual - Before and after each use and after cleaning	Recommendations for the implementation of a respiratory protection program are outlined in *Section 2* of AS/NZS 1715 Maintenance requirements outlined in *Section 9* of AS/NZS 1715 – in particular *Table 9.1* specific storage requirements other than those listed in 5.4.	AS/NZS 1715 AS/NZS 1716	Training requirements outlined in *Section 7* of AS/NZS 1715 Fit tests and check requirements outlined in *Section 8* of AS/NZS 1715

PPE Type	Selection	Inspection Frequency	Maintenance and Storage Requirements	Applicable legislation, Codes of Practice or Standards	Minimum Training or Testing Requirements
Fall Protection					
Individual fall arrest systems and devices	Selection recommendations outlined in *Section 2* of AS/NZS 1891.4	Inspection frequencies outlined in *Section 9* of AS/NZS 1891.4	As per manufacturer's instructions No specific storage requirements other than those listed in 5.4.	AS/NZS 1891.1 AS/NZS 1891.4	Testing - Schedule of tests for fittings, components and assemblies outlined in *Table 1* of AS/NZS 1891.1 Training – Guidance on the provision of training is outlined in *Appendix E* of AS/NZS 1891.4

جامعة خليفة
Khalifa University

Record of Reviews & Revisions

REV.	DATE ISSUED	DESCRIPTION OF REVISION		REVIEWED BY	AUTHORIZED BY
		PAGE NO	COMMENT		
01	07/06/2018	All	Change of logo	Roberto M.	Belal I.
		Throughout	Change from KUSTAR to KU	Roberto M.	Belal I.
		Throughout	Change the reference from AD EHSMS to OSHAD SF	Roberto M.	Belal I.
		02	Prepared by Reviewed / Updated by, Endorsed for Approval by, and Approval & Issue by	Roberto M.	Belal I.
		21	Change the reference from AD EHS Center to OSHAD	Roberto M.	Belal I.
		21	Reference Standard from ISO 14001:2004 to ISO 14001:2015	Roberto M.	Belal I.
		21	Reference Standard from OHSAS 18001 2007 to ISO 45001:2018	Roberto M.	Belal I.

Copies of this document available in KU EHS Intranet are only the latest, valid and controlled documents. All other copies are uncontrolled and valid only on the date printed. Last printed: 15/6/2020 3:58:00 PM